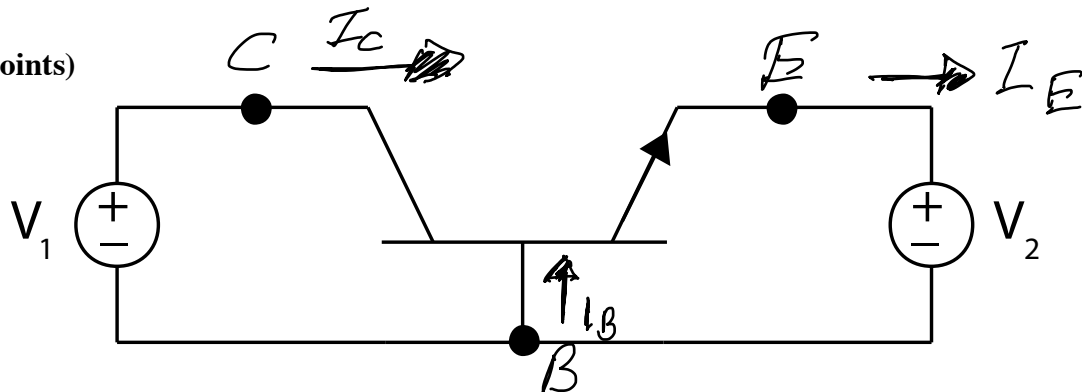


GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

ECE 3040 Practice Test #2

Name: _____

1. (12 points)



(a) (2 points) What type of BJT is the transistor shown above? (NPN or PNP)

NPN

(b) (2 points) Identify emitter, base and collector terminals and clearly show on the figure above.

See figure

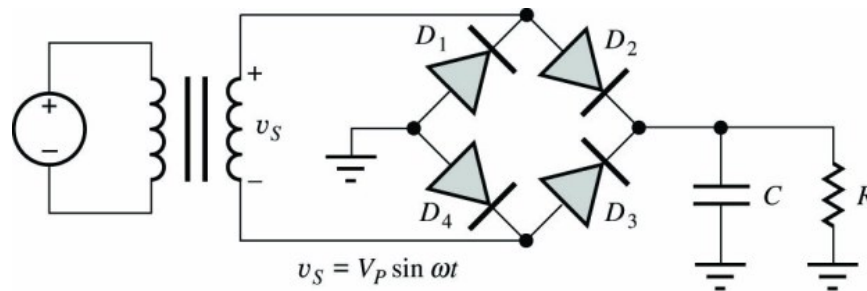
(c) (4 points) Determine the mode of operation under following conditions:

V_1	V_2	Mode of Operation
Positive	Positive	<i>Cut-off</i>
Positive	Negative	<i>Active</i>
Negative	Positive	<i>Inverted</i>
Negative	Negative	<i>Saturation</i>

(d) (4 points) Clearly show the **direction of current flow** at the emitter, base and collector terminals on the figure above when the transistor is operated in the **forward active mode**.

See figure

2. (6 points) In the rectifier circuit given below,



- (a) (2 points) Which diode(s) conduct in negative cycles?

D_1 and D_3

- (b) (2 points) Turn-on voltage for each diode in the circuit is V_{on} . What is the output DC voltage?

$$V_p - 2V_{on}$$

- (c) (2 points) How can you reduce the ripple voltage of this rectifier circuit?

Increase C or R

Increase source frequency.

3. (4 points) Why is the current gain different under forward active and inverted biasing of a BJT? Which of the two biasing conditions leads to higher current gain and why?

In inverted mode, emitter efficiency is lower due to lower doping concentration of the collector.

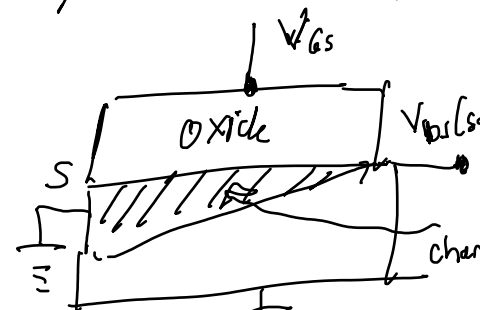
Active mode leads to higher current gain

4. (4 points) Explain the base width modulation (Early effect) in a BJT? What small signal parameter does it affect?

The voltage bias on the p-n junctions in a BJT leads to changes in the depletion layer widths. This changes the effective base width. Early Effect is captured with r_o in the small signal model.

5. (4 points) Explain the pinch-off point in a MOSFET. Under what conditions does it happen? You can support your explanation with a sketch.

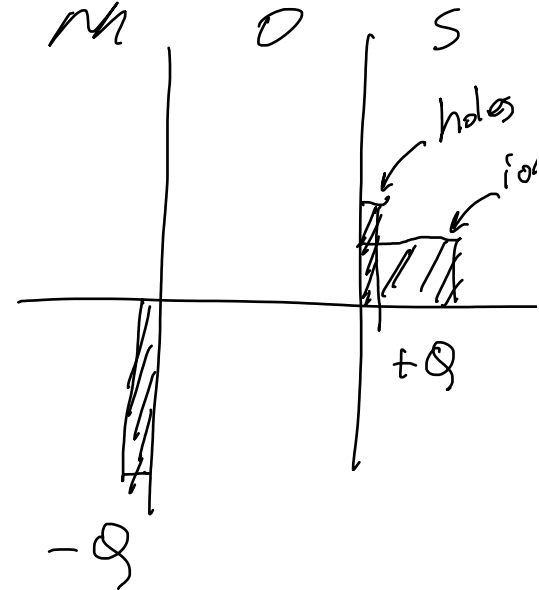
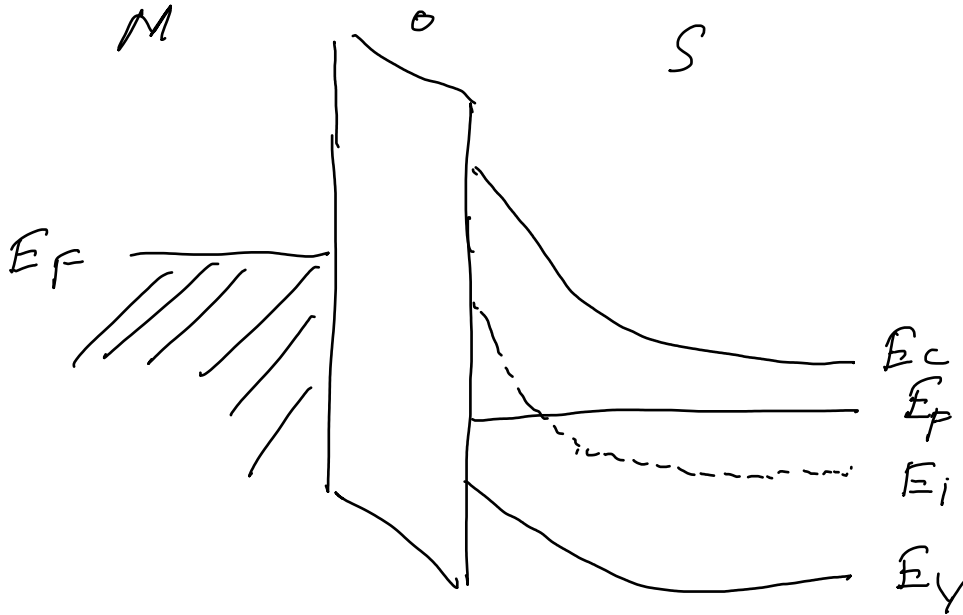
When $V_D = V_{Dsat} = V_G - V_T$, the drain end of the inversion layer is at the onset of inversion. The channel pinches off because of this.



6. (10 points) Sketch the energy band diagram and charge diagram of an ideal MOS structure for the following conditions. Also indicate the biasing regime for each case.

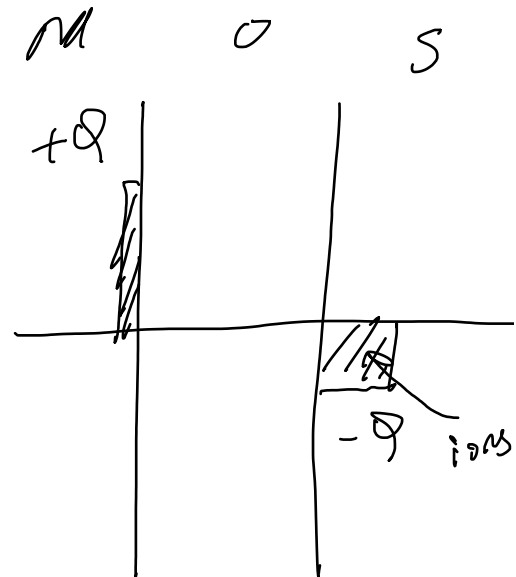
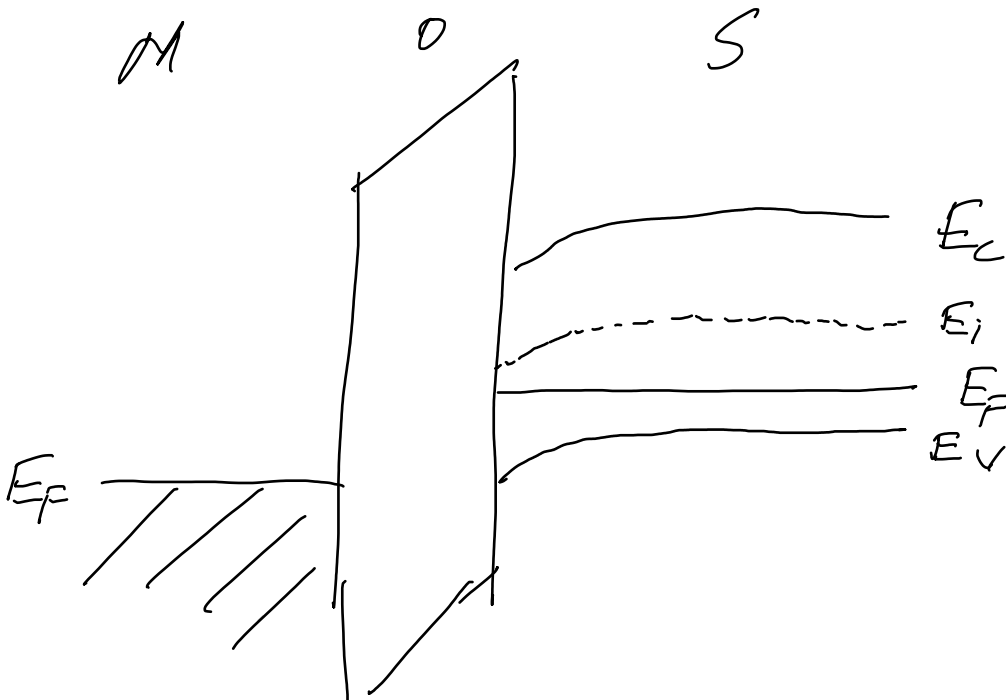
(a) (5 points) $\phi_F = -12kT/q$, $\phi_S = -26kT/q$

Inversion



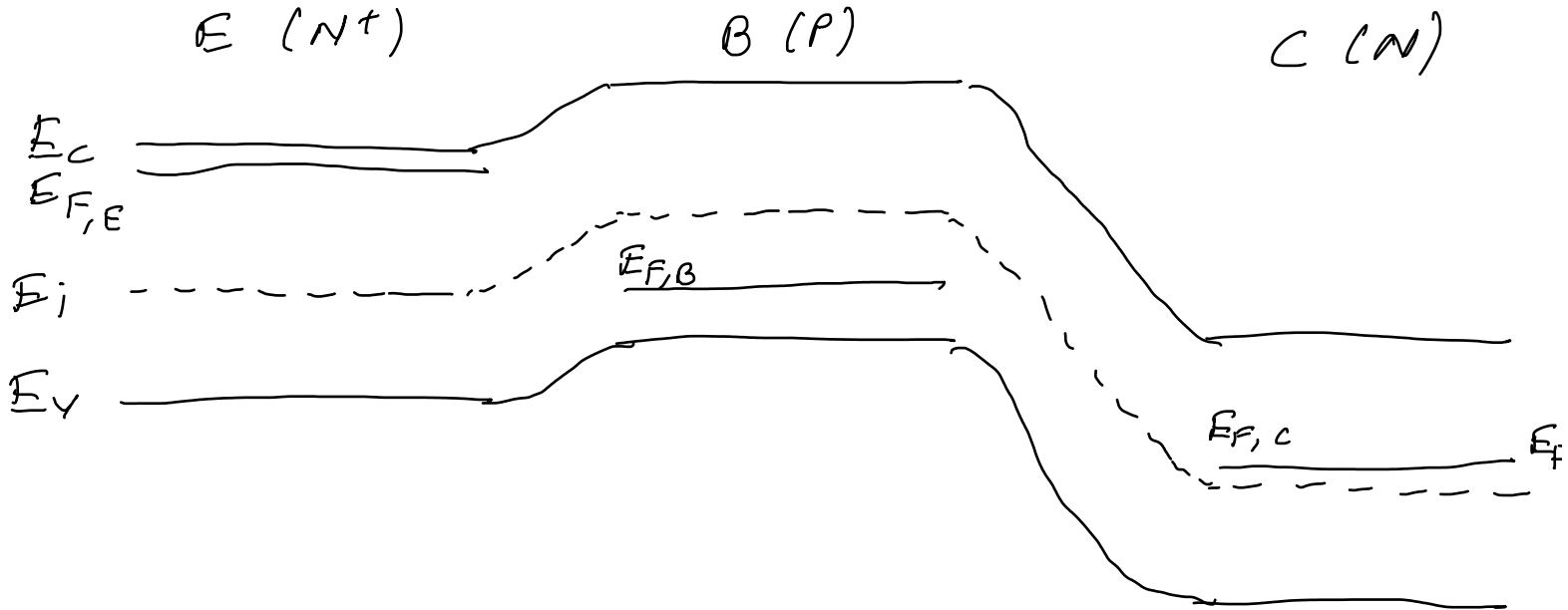
(b) (5 points) $\phi_F = 9kT/q$, $\phi_S = 6kT/q$

Depletion

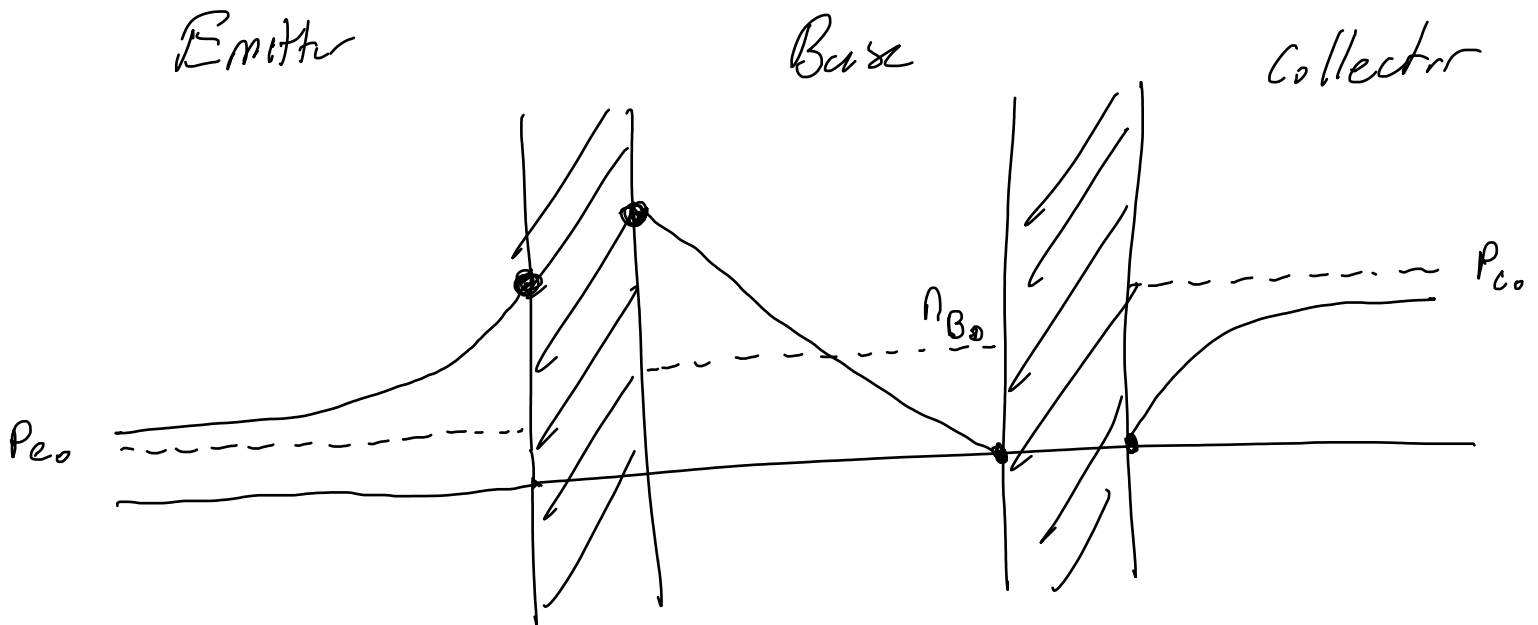


7. (10 points) Consider a NPN BJT with doping densities $N_E \gg N_B > N_C$.

(a) (5 points) Sketch the energy band diagram under **forward active mode biasing**. Clearly indicate emitter, base and collector regions and label E_C , E_V , E_i and E_F .



(b) (5 points) Roughly sketch the **total minority carrier** concentration in the emitter, base and collector regions under **forward active mode bias**. Please indicate the minority carrier type and equilibrium concentrations, mark the emitter, base and collector in your sketch, and assume $W/L_B \ll 1$.



Problem 1 (10 points): Consider a BJT with the doping concentrations for the emitter, base and collector are 10^{18} , 10^{16} , and $5 \cdot 10^{15} \text{ cm}^{-3}$, respectively. The base width is $2 \text{ } \mu\text{m}$, the diffusion lengths of the minority carriers are $L_E = L_C = 6 \text{ } \mu\text{m}$ and $L_B = 20 \text{ } \mu\text{m}$ and the diffusion coefficients are $D_E = D_B = D_C = 5 \text{ cm}^2/\text{s}$.

(a) (5 points) Calculate the emitter efficiency, the base transport factor, and the common emitter current gain of the transistor in **active** mode assuming $W \ll L_B$.

$$\alpha_T = \frac{1}{1 + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} = \frac{1}{1 + \frac{1}{2} \left(\frac{2}{20} \right)^2} = 0.995$$

$$\gamma = \frac{1}{1 + \left(\frac{D_E}{D_B} \frac{W}{L_E} \frac{N_B}{N_E} \right)} = \frac{1}{1 + \left(\frac{5}{5} \frac{2}{6} \frac{10^{16}}{10^{18}} \right)} = 0.9967$$

$$\beta_o = \frac{1}{\left(\frac{D_E}{D_B} \frac{W}{L_E} \frac{N_B}{N_E} \right) + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} = \frac{1}{\left(\frac{5}{5} \frac{2}{6} \frac{1}{100} \right) + \frac{1}{2} \left(\frac{2}{20} \right)^2} = 120$$

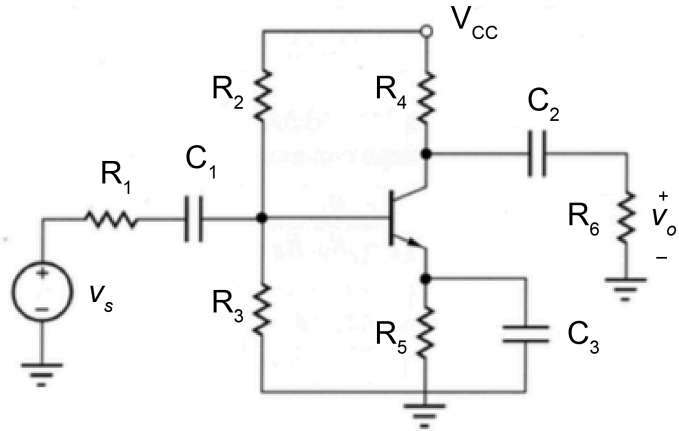
(b) (5 points) Calculate the emitter efficiency, the base transport factor, and the common emitter current gain of the transistor in **inverse** mode assuming $W \ll L_B$.

$$\alpha_{TR} = \frac{1}{1 + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} = 0.995$$

$$\gamma = \frac{1}{1 + \left(\frac{D_C}{D_B} \frac{W}{L_C} \frac{N_B}{N_C} \right)} = \frac{1}{1 + \left(\frac{5}{5} \cdot \frac{2}{6} \cdot \frac{10^{16}}{5 \times 10^{15}} \right)} = 0.6$$

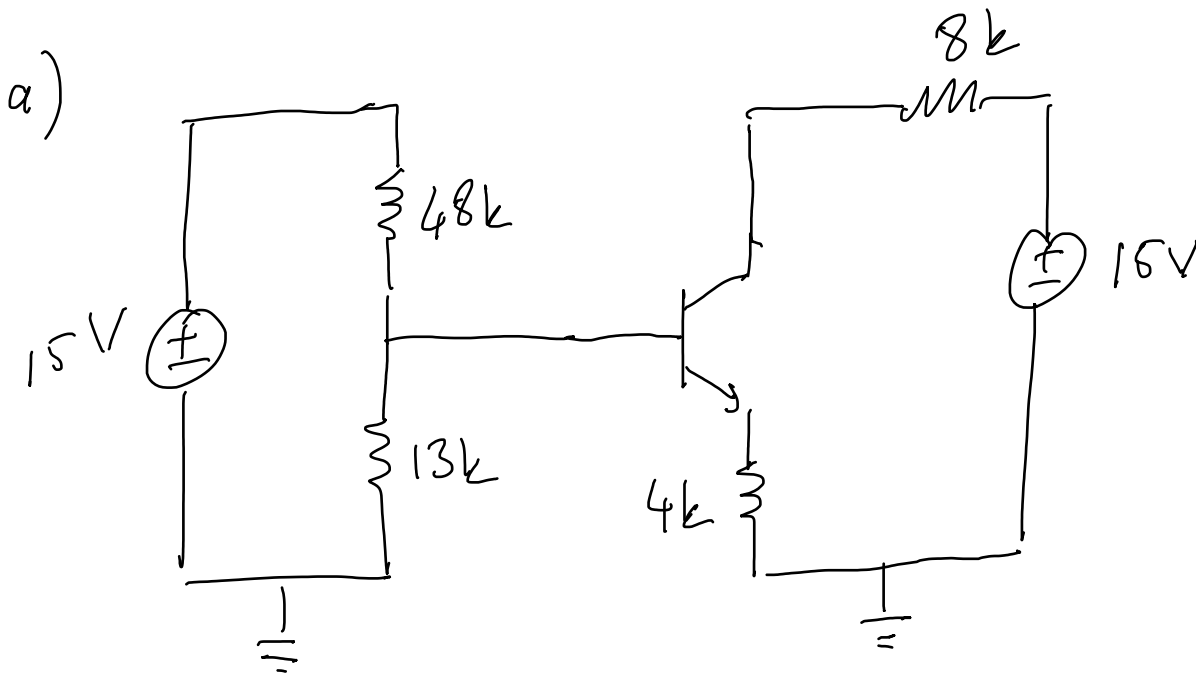
$$\beta_R = \frac{1}{\left(\frac{D_C}{D_B} \frac{W}{L_C} \frac{N_B}{N_C} \right) + \frac{1}{2} \left(\frac{W}{L_B} \right)^2} = \frac{1}{\left(\frac{5}{5} \cdot \frac{2}{6} \cdot \frac{10^{16}}{5 \times 10^{15}} \right) + \frac{1}{2} \left(\frac{2}{20} \right)^2} \approx 1.41$$

Problem 2 (40 points): For the circuit shown on the right, assume the transistor is operated in the active mode, $\beta_F = 100$, $V_{ON} = 0.7 \text{ V}$, $V_A = 100 \text{ V}$, $R_1 = 1 \text{ k}\Omega$, $R_2 = 48 \text{ k}\Omega$, $R_3 = 13 \text{ k}\Omega$, $R_4 = 8 \text{ k}\Omega$, $R_5 = 4 \text{ k}\Omega$, $R_6 = 150 \text{ k}\Omega$ and $V_{CC} = 15 \text{ V}$.



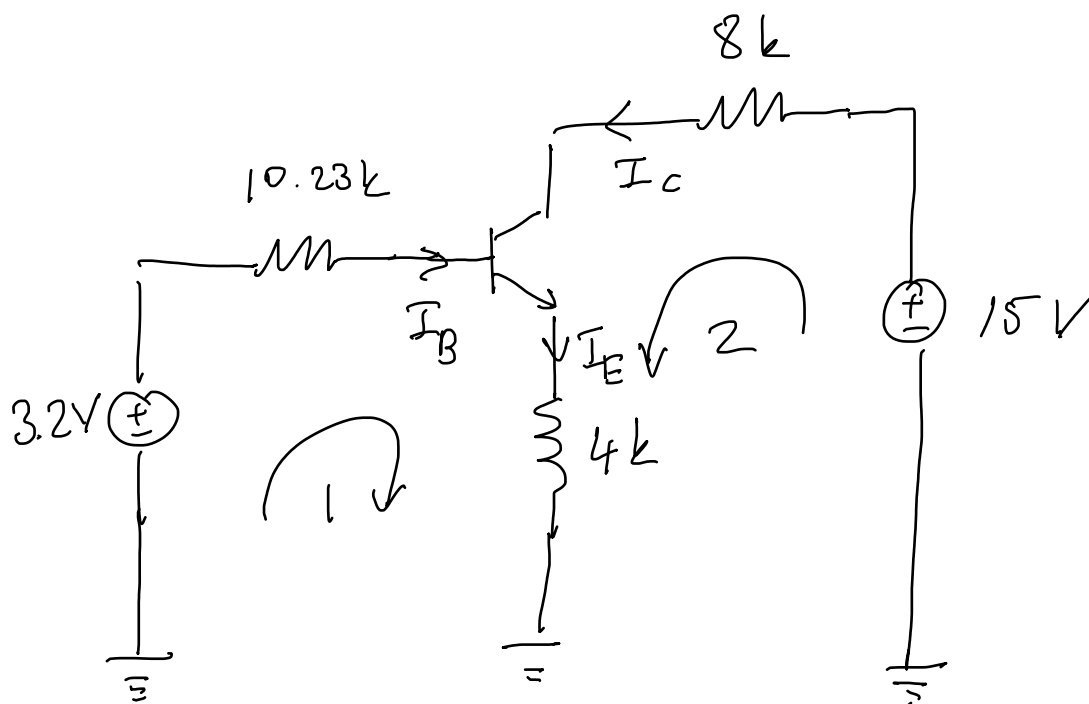
(a) (20 points) Calculate the Q-point, i.e. (I_C , V_{CE}), of the BJT. Hint: Draw the DC equivalent circuit (5 points). First find the Thevenin equivalent of the base bias circuit (5 points) and then calculate the transistor Q-point (10 points).

(b) (20 points) Calculate the signal source voltage gain v_o/v_s of the circuit. Hint: Draw the AC equivalent circuit and simplify the circuit by calculating its Thevenin equivalent (5 points). Calculate the small signal model parameters and replace the transistor with its small signal equivalent circuit (10 points). Then calculate the voltage gain of the circuit (5 points).



$$V_{th} = \frac{13k}{48k + 13k} \cdot V_{CC} = 3.2 \text{ V}$$

$$R_{th} = 48k // 13k = 10.23 \text{ k}\Omega$$



Loop 1 : $3.2 = 10.23k I_B + \overset{0.7V}{V_{BE}} + 4k I_E$

$$2.5 = (10.23k + 101 \times 4k) I_B$$

$$\Rightarrow I_B = 6.04 \mu A$$

$$I_C = 604 \mu A$$

$$I_E = 610 \mu A$$

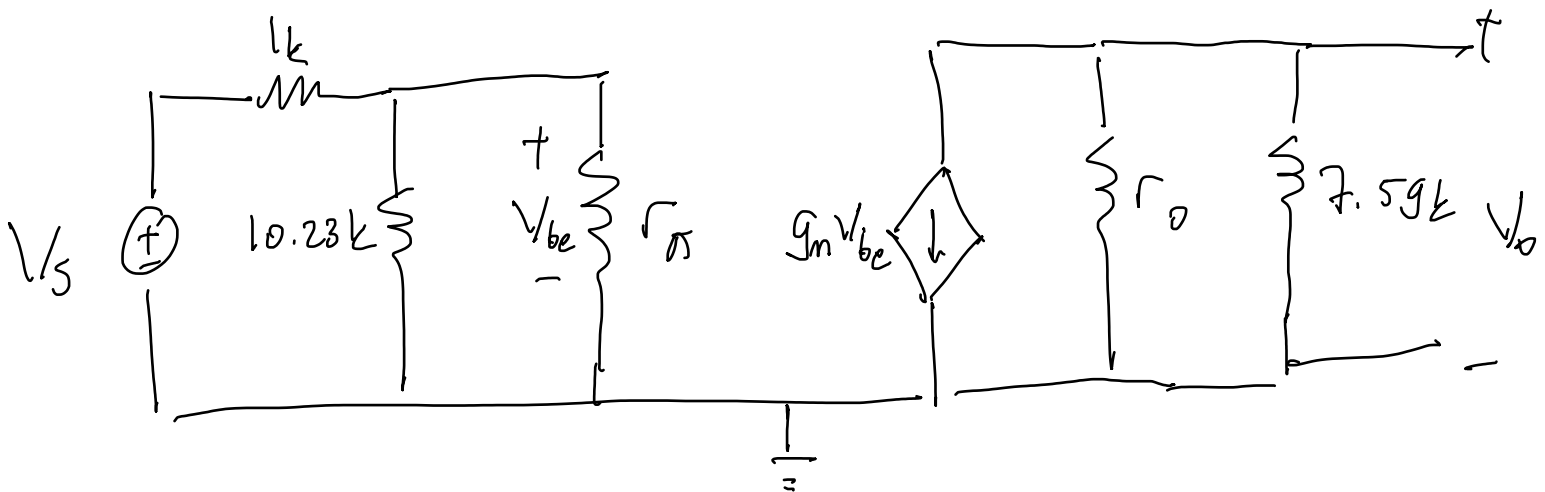
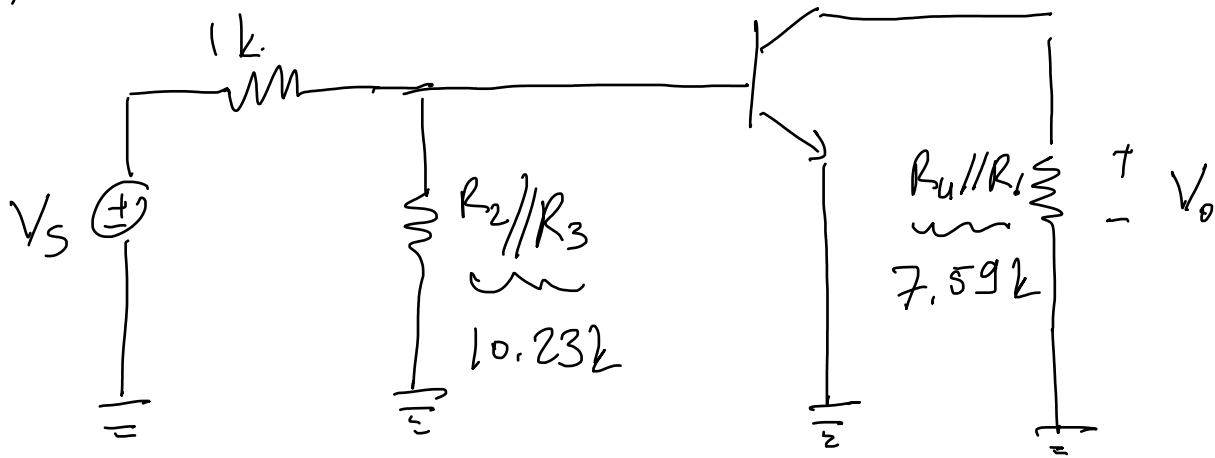
Loop 2: $15 = 8k I_C + V_{CE} + 4k I_E$

$$\Rightarrow V_{CE} = 15 - 8k I_C - 4k I_E = \boxed{7.728V}$$

$$Q_{point} (I_C, V_{CE}) = (604 \mu A, 7.728V)$$

Forward active ✓

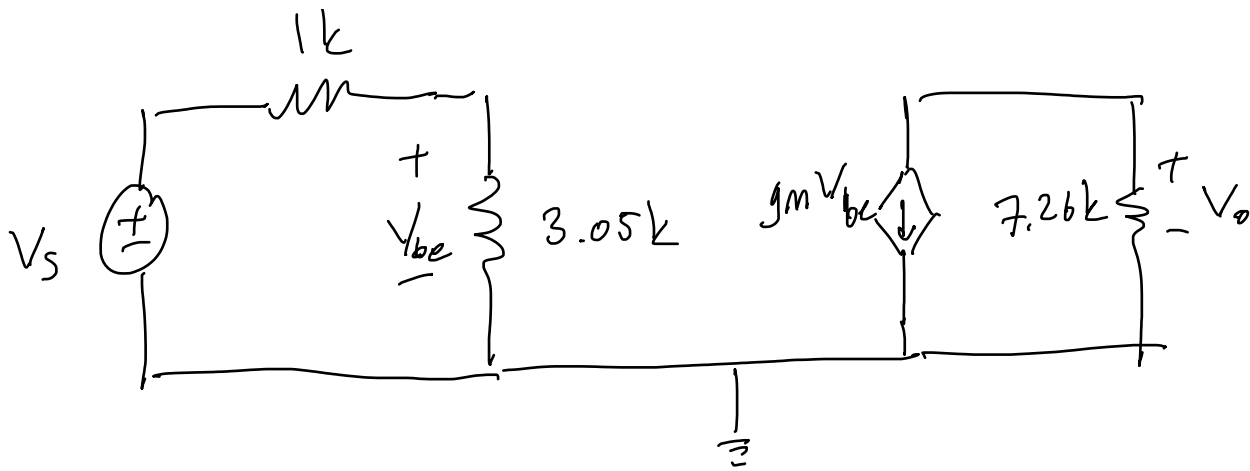
(6)



$$g_m = \frac{q I_c}{kT} = \frac{604 \mu A}{0.0259} = 0.023 S$$

$$r_{\pi} = \frac{\beta_o}{g_m} = \frac{100}{0.023} \approx 4.35 k\Omega$$

$$r_o = \frac{V_A}{I_c} = \frac{100V}{604 \mu A} = 165.56 k\Omega$$



$$V_{be} = \frac{3.05k}{3.05k + 1k} \cdot V_S = 0.753 V_S$$

$$V_o = -g_m V_{be} \times 7.26k$$

$$= -0.023 \times 0.753 V_S \times 7.26k$$

$$= -125.8 V_S$$

$$A_v = - \frac{125.8 \cancel{V_S}}{\cancel{V_S}} = \boxed{-125.8}$$

